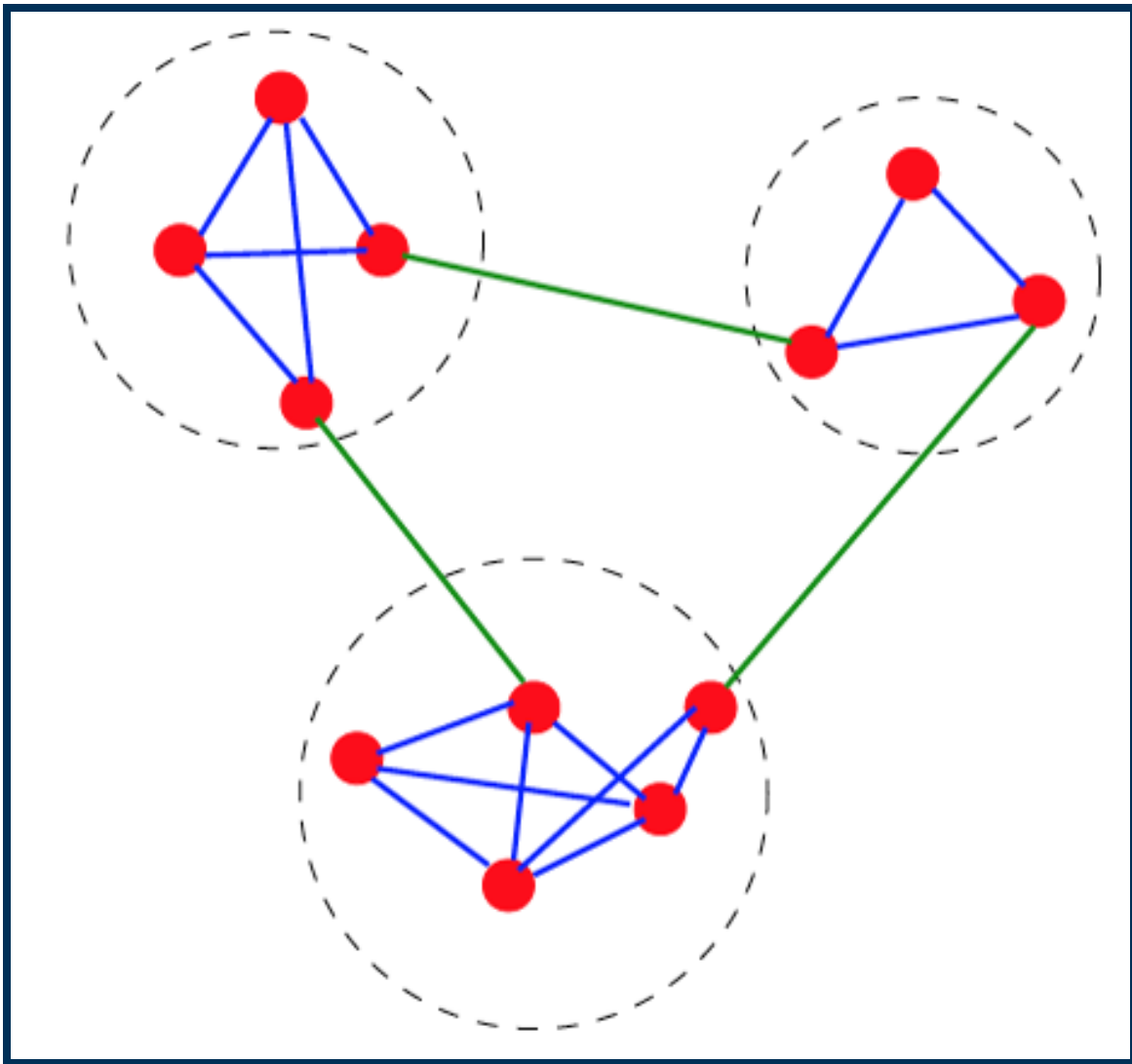
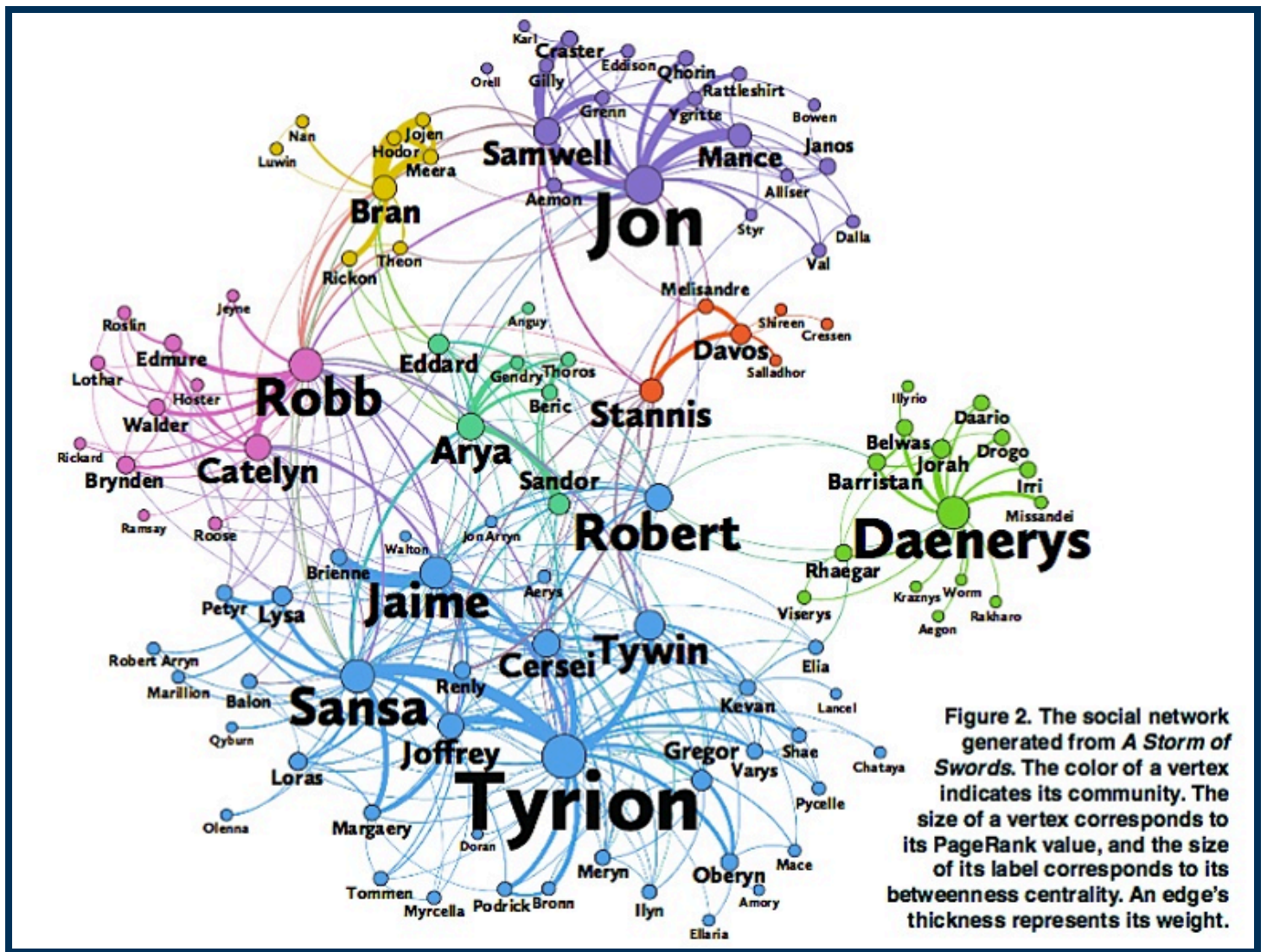


L1: Communities (Modules) in Networks



Another important problem in Network Science is to discover Communities – in other words, clusters of highly interconnected nodes. The density of the connections between nodes of the same community is much larger than the density of the connections between nodes of different communities.



Returning to the previous Game of Thrones visualization, each color represents a different community – with a total of 7 communities of different sizes.

For those of you that are familiar with the book or TV show (*mostly seasons 3 and 4*), these communities make a lot of sense. Up to that point in the story, Daenerys, for instance, was mostly interacting with the Dothrakis and with Barristan, while Jon was mostly interacting with characters at Castle Black.

There are many algorithms for Community Detection – and some of them are able to identify nodes that participate in more than one community. We will discuss such algorithms later in the course.